HARSHITA BOSE

B.Tech. Engineering Physics | Delhi Technological University
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RESEARCH INTERESTS

Integrated quantum photonics; semiconductor quantum devices; quantum emitters and dots; nanophotonics; photonic-device simulation and fabrication; quantum photonic waveguides

EDUCATION

Delhi Technological University 2023–2027
- B.Tech. in Engineering Physics | CGPA: 9.44/10.0 | Department Rank: 3/120

Central Board of Secondary Education (CBSE) 2023
- Overall score: 93%

Delhi Public School – R. K. Puram 2009–2023
- Gold Medal for Academic Excellence; Scholar Badge Awardee for 7 consecutive years, Blue Blazer Awardee

RESEARCH EXPERIENCE

Integrated Quantum Photonics Lab, Humboldt-Universität zu Berlin | ***Intern*** **May–Aug 2026**
- Simulated and characterized integrated quantum-photonic waveguides using JCMwave and Python.
- Designed and optimized S-bends and U-bends for quantum communication applications.
- Internship supported by Santander Bank through the Humboldt Internship Program.

QTRAN Lab, IISER Thiruvananthapuram | ***Intern*** **Jun–Jul 2025**
- Designed and implemented a quantum router based on three quantum dots; studied wavefunction dynamics and confinement quantization, with results validated through experimental measurements. *Manuscript in preparation.*
- Worked on PCB design and fabrication, chemical etching, soldering, wafer-to-PCB wire bonding, and variable-capacitor design.

Nanoscale Research Facility (NRF), IIT Delhi | ***Semiconductor Device Fabrication Training*** **Jul 2025**
- Fabricated semiconductor devices in a Class 100/1000 cleanroom, including wafer cleaning, oxide-film formation, and device lift-off.
- Hands-on experience with spin coating, maskless photolithography, and titanium deposition by e-beam evaporation.

Space Systems Lab, IIIT Delhi | ***Intern*** **Jun–Jul 2024 and Dec 2023**
- Evaluated Monte Carlo, Patera, Alfano, and Advanced Extrapolated State Propagation techniques for LEO/MEO satellites.
- Proposed a Cluster-based ESPT incorporating machine-learning techniques to improve propagation performance.

UNDERGRADUATE RESEARCH PROJECT | DTU–IIT Delhi

- **Numerical Modeling and Optimization of High-Fidelity Spin Shuttling in Silicon Quantum Dots**

PUBLICATIONS

- **H. Bose**, "Electrical Selection of Two Integrated Quantum Emitters using Pulse-Programmable Triple Quantum Dots," accepted for presentation at the *17th International Conference on Fiber Optics & Photonics (PHOTONICS 2026)*, IIT Hyderabad, 9–13 Dec. 2026.
- R. Bharadwaj, **H. Bose**, and M. Thalakulam, "A resonant-tunneling quantum-dot demultiplexer for scalable cryogenic single-electron routing," *Journal of Physics: Condensed Matter*, submitted for publication, Oct. 2026.
- **H. Bose**, "Pulse-Programmable Processing in Coulomb-Blockaded Triple Quantum Dots for Hybrid Quantum Optical Architectures," presented at the *XLIX OSI Symposium, Frontiers in Optics and Photonics (FOP) 2026*, New Delhi, India, 25–28 Sept. 2026.
- **H. Bose**, "Transit Photometry for Estimating the Velocity of Exoplanets and Specific Defence Applications," in *proc. 2024 IEEE Space, Aerospace and Defence Conference (SPACE)*, Bangalore, India, 2024, pp. 450–453. DOI: 10.1109/SPACE63117.2024.10667777.

TECHNICAL SKILLS

- **Programming & Data:** Python, MATLAB, C/C++, LaTeX
- **Quantum/Photonic Simulation:** JCMwave, Kwant, COMSOL, Qucs, KLayout
- **Fabrication & Electronics:** PCB design, wafer microfabrication, photolithography, e-beam evaporation, wire bonding
- **Physics:** Quantum mechanics, advanced mathematics and physics, semiconductor devices
- **Other:** Machine learning (CNN, reinforcement learning), AI for materials science, Digital/Analog Electronics, Arduino, Raspberry Pi

AWARDS & HONORS

- **Humboldt Internship Program Scholarship —** Humboldt-Universität zu Berlin; funded by Santander Bank, May–Aug 2026
- **Taiwan National Science and Technology Council (NSTC) International Internship Pilot Program (IIPP) —** Selected, May 2026
- **ThinkSwiss Research Scholarship 2025 —** Shortlisted; May 2025
- **Invictus Hackathon 2024 —** 2nd Prize, Feb 2024
- **Indian National Space Settlement Design Competition —** Qualifier; Team Aurora, Aeross, 2021
- **Hindustan Times Open Essay Competition —** Winner, 2015

LEADERSHIP, ACTIVITIES & SERVICE

IEEE | *Student Member* **Aug 2024–May 2026**

- Mentor, Electronics and Embedded Systems, Techweek 2025; delivered a motivational talk to first-year students.

HEIV – The Society for Popularization of Science | *Member* **Feb 2024–May 2026**

- Supported science communication and public outreach; participated in recruitment and expansion initiatives.

CANSAT DTU Institute Team | *Member* **Sep 2023–May 2024**

- Contributed to a can-sized satellite design challenge organized internationally by the American Astronautical Society and locally by ISRO; worked on descent, stabilization, and navigation.

Feeding Hope | *Volunteer* **Mar–Aug 2021**

- Supported fundraising for ration distribution during COVID-19.

Music & Sports: Hindustani classical music (3rd year); Rabindra Sangeet and harmonium; lawn tennis; swimming.

LANGUAGES

English — Proficient | Hindi — Proficient | Bengali — Proficient | German — Basic
TOEFL iBT — 6/6 (118/120), CEFR C2